HACK-O-MED 2023

Round 2

Team Name : Plast-o-Free Med

Team Details & Problem Statement

Team ID : HMED_A15

Track : Bio-Waste Management

Problem Statement : A new prospect for Recycling of Plastic waste generated from Bio-medical facilities

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ABOUT US

Team Name :

PlastOfree Med

>> Initial thought >>

Our proposed prospect for Recycling of Medical Plastic waste was firstly a rough idea brought into reality with our technical team members under the guidance of our Technical Project Mentor in our college Institute Of Engineering & Management, Kolkata..

<< Medical intervention <<

Later on as we built the prototype and needed Medical informative Guidance for perfecting the Recycling-process we joined hands with Burdwan Medical College and took the mentorship of a dignified Medical Professor along with two peer aged Medical students who showed similar concern for this rising problem of Plastic pollution and wanted to contribute on behalf of the Medical community.

>> Teaming up together >>

It is our similar Environmental concern and problem identification which made us collectively work towards for building a solution for a better and less plastic-polluted future.

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CONTENT KEY...

1. Brief Introduction : What is Plastic Med-Waste Recycler ?
2. Defining the Problem : Stating Causes & Source of Medical Plastic-waste pollution
3. Our Solution : Proposal, Implementation & Cost Structure
4. Detailed Process of Recycling : Step-by-step
5. Profound Impact : bringing change to current scenario
6. The Team of Plast-O-Free Med : team Intro

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1. BRIEF INTRODUCTION

Plastic Med-Waste Recycler (PMWR)

Recyclable waste

PMWR has been developed to recycle the huge number of syringes and other waste materials such as medical fluid bottles, plastic medical instruments and other tubular plastic-waste which are discarded after single-use.

With the aid of its adjustable sliding mechanism, PMWR accepts various kinds & sizes of Solid tubular plastic-waste.

Sanitization has been taken care with utmost importance in PMWR for which it has a specially sealed container isolating waste from air of outside atmosphere and also has a simple cleansing mechanism for sterilizing the waste against any kind of health-hazard.

PMWR individually recycles each plastic waste into plastic-beads of cylindrically-shaped 5mm size.

PMWR requires very minimum maintenance and moderately less human-intervention as it's easy to switch & control and very user-friendly, without any manual handling of plastics by the operator.

PMWR generates revenue stream since manufactured plastic-beads would be upto manufacturing-grade and sellable to the market.

PMWR's Process & Technology stack

1. Mass Collection & Isolation
2. Melting section
3. Cooling Section
4. Cutting Section
5. Collection system

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2. DEFINING THE PROBLEM

Problem Statement : A new prospect for Recycling of Plastic waste generated from Bio-medical facilities

>> Cause <<

Plastic-pollution has always been a great problem posing a major threat to environment being a non-biodegradable type of waste. Among Plastic waste, Bio-medical wastes also contribute a considerable amount. If seen from the point of view of hospital authority, Plastic syringes and packaging containers are to necessary hospital-logistics. Plastic is even used as raw material in several medical equipment whether surgical or not. Medical waste product needs to be recycled for ultimately stopping plastic pollution generated due to accumulation of untreated bio-medical waste.

Reasons for Recycling

- Reduction in single-use Medical plastic-waste
- Lowering Medical-plastic instruments cost
- Conservation of resources and energy
- Safeguard Environment from Incinerated pollutants

Sources Of Medical-waste

- In Diagnostic Centre : here a lot of syringes and sample collection / testing tubes are discarded for collecting & processing blood samples.
- At Health Centre : huge number of syringes discarded during vaccinations and also medicine-containers made of hard recyclable plastic.
- At Hospitals : All types of Plastic medical fluid-containers, Syringes and other tubular waste-products.

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3. OUR SOLUTION

Solution to the discussed Problem Statement has been arrived at with our invention, Plastic Med-Waste Recycler.

SOLUTION

PROPOSAL

- Of The total amount of waste generated by health-care industry, about 85% is general non-hazardous waste recyclable in nature.
- The unique value of Plastic Med-waste Recycler proposes that it recycles bio-medical plastic-waste making the environment free of non-biodegradable waste by producing a raw-material that can be used in the manufacturing of its own kind.
- Hence, it's important to recycle the medical plastic waste products through a definite process.

IMPLEMENTATION

- INSTALLATION :

1. At Hospitals , Nursing Home : (Governement & Private both)
2. At Blood and Health Diagnostic Centres : (e.g., Suraksha Diagnostics, etc.)
3. At Medical health Centres : (e.g. Swasthya Kendras, vaccination Centres)

- OPERATION :

Once a week / daily recycling of collected plastic-waste and packeting fresh Plastic-beads.

- COLLECTION :

Selling of Plastic beads by weight at Market-rate by operators themselves or some raw-material supplier.

COST-STRUCTURE

- Initial Cost Of Building Plastic Med-Waste Recycler : ₹ 6000.00 /-
- Projected Cost Of Manufacturing Plastic Med-Waste Recycler : ₹4800.00 /-
- Operations Cost : Electricity consumption cost : ₹ 780 .10 / (@ ₹ 8.90/ kWh*2hrs/day*1yr) Hospital Disinfectants : Inventory
- Revenue : Market-price of Plastic-granules : ₹ 40.00 to ₹ 100.00/kg
- Selling amount : 10kg beads (in 1 month from an Hospital)
- Cost recovery period : 14 months

*All figures estimated on lower limits

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4. DETAILED PROCESS OF RECYCLING

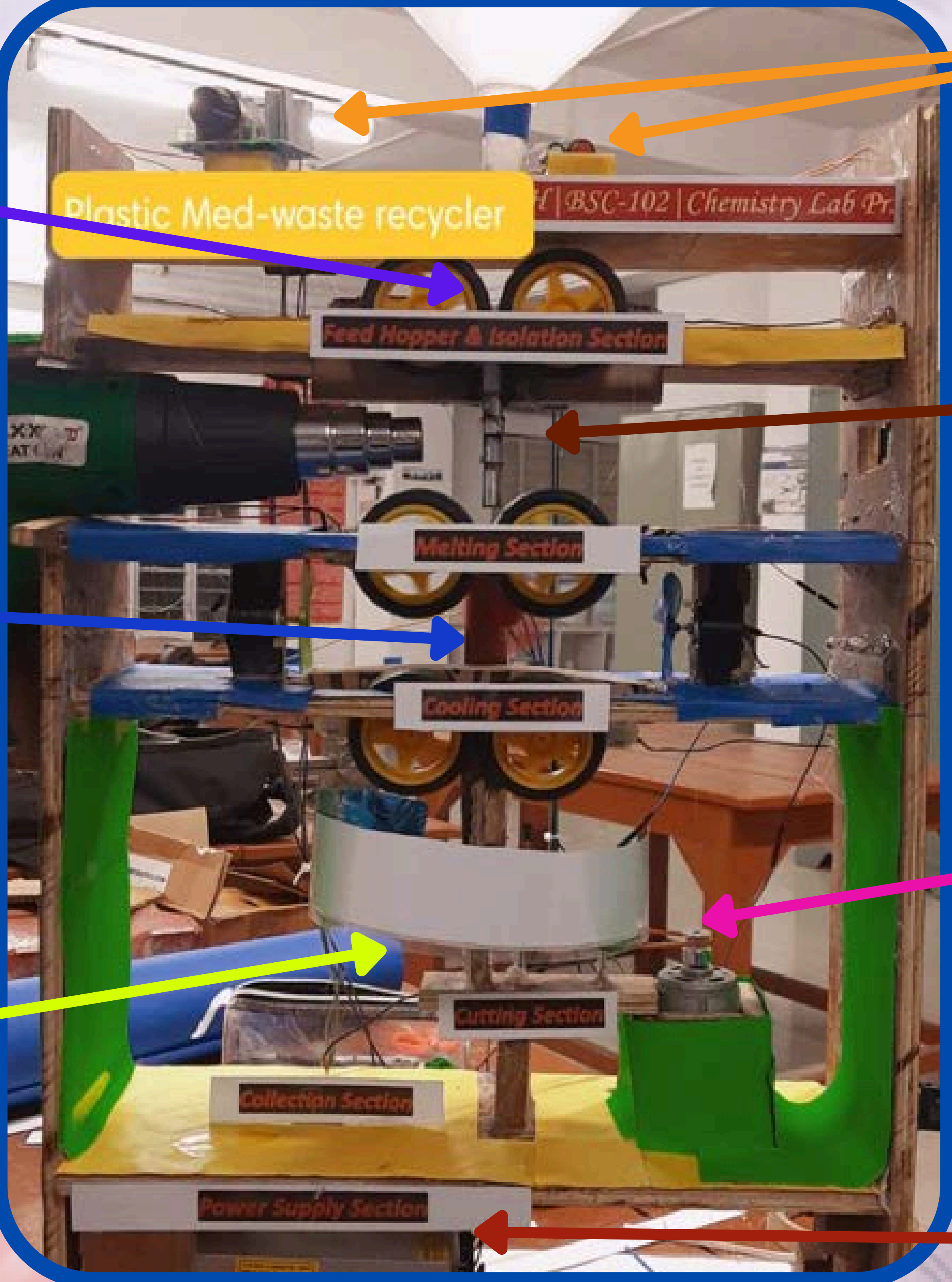
STEP-I :

MASS COLLECTION & ISOLATION SECTION >>

The disposed Medical plastic-waste is first sanitized and sterilized by dipping in a Cleansing Solution and then disposed off into the Recycler wherein one waste is rolled-in at a time.

<< CONTROL SECTION

Two DC Speed Controls are fixed atop of the recycler to manually control the speed of the Wheels and the Cutting-blade.



STEP-II :

<< MELTING SECTION

After being rolled-in by the isolator-wheels it is sent down in front of a High-temperature Hot-Air Gun to make the plastic-waste reach a semi-molten state.

STEP-III :

COOLING SECTION >>

Since hot semi-molten plastic can't be processed directly hence the hot strands are pressed by a pair of wheels and cooled by 3 fans down to room-temperature.

STEP-IV :

<< CUTTING SECTION

After getting cooled it is being again pressed by a pair of wheels to eliminate any inner hollowness and sent down to a high-speed rotating blade powered by a high-speed high-torque DC motor.

STEP-V :

COLLECTION SECTION >>

After being cut by the high-speed rotating blade into small plastic-beads / granules they fall into the collection tray, thus completing the process.

<< POWER-SUPPLY SECTION

The Recycler receives Power from the SMPS fixed at the bottom which takes in AC from Mains Supply and gives DC power to each part.

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5. PROFOUND IMPACT

Plastic Med-Waste Recycler (PMWR)

With this project "Plastic Med-Waste Recycler", we can reduce plastic pollution in our environment. In today's world plastic pollution is really a very concerning threat for the world. We have made this project with a view to reduce the medical based plastic waste products like syringes, saline bottle, plastic containers which is also harmful for the society if it is not sanitized properly before disposing.

Recently for Covid-19 vaccination process, a lot of plastic syringe waste has generated which needs to be recycled properly otherwise it creates a severe pollution in the nature. So, it is the high time to think about the problem and to generate a solution for this.

First and foremost, it significantly reduces the environmental impact of healthcare activities. Often the Medical plastics which are used are single-use plastics (like syringes) and packaging, contributing to the growing problem of plastic pollution.

Secondly, recycling medical plastic waste conserves valuable resources and prevents emissions from harmful Incinerated pollutants. Plastic production requires fossil fuels and energy-intensive processes, contributing to greenhouse gas emissions and depletion of non-renewable resources.

Reusing recycled materials from medical plastic-waste itself reduces the need to manufacture new plastics, saving on raw material expenses.

REDUCE REUSE RECYCLE

Medical Waste Statistics

1. Daily Plastic Medical waste generated from Hospitals worldwide: 1.7 Million Tons/day
2. 16 Billion/year injections administered worldwide
3. 44 Lakh/day Syringe-waste produced worldwide
4. 3.1 Metric Tons Of Plastic-beads can be manufactured daily from syringe-waste produced worldwide
5. 150Cr Syringes needed to vaccinate whole Indian population 220Cr Vaccines administered in actual in India till date

6. TEAM OF PLAST-O-FREE MED

Suvojit Maity,
Team Co-Lead

Shyantan Kundu,
Team Lead

Adrija Ghosh,
Technical Member

Subhradip Bal,
Technical Member

Prof. Dr. Chandan Adhikari ,
Technical Project Mentor

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Thank you very much!